Understanding host-pathogen interactions in patients with syphilis infection

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Background & Rationale

Syphilis continues to cause substantial morbidity and mortality worldwide. The World Health Organization has estimated that there are over 6 million new cases of syphilis globally each year, and incidence has more than doubled in the United States since 2000. Currently, the diagnosis of syphilis relies on immunological determinants first described more than 100 years ago (i.e. reagin test) and the available prevention interventions and clinical management protocols for syphilis have not been substantially improved in more than 75 years. Further research is urgently needed to advance the understanding of syphilis pathogenesis and improve syphilis control, including an improved understanding of the immune-biology of syphilis infection for vaccine development and enhanced detection methods through rapid point-of-care diagnostics.

In 2012, as part of an NIH research capacity development program (NIH/NIAID 5R01AI09972), our group initiated a longitudinal cohort study of syphilis in Peru (the "PICASSO study") of over 400 men and transgender women. We built upon 13 years of prior UCLA and USC-Peru collaborations in epidemiological, clinical and laboratory-based research (U01MH61536 NIMH CPOL and R01MH078752 NIMH CPOS). In PICASSO through 2017 we successfully 1) enrolled a cohort of 401 people, retaining 77% of them over 24 months; 2) strengthened the capacity of our sexual health laboratory to conduct molecular biology research including *Treponema pallidum* (TP) genotyping, molecular antimicrobial TP resistance determination, and various TP enzyme immunoassays; 3) created a biospecimen repository of over 3000 well-characterized specimens; 4) conducted evaluations of new point-of-care rapid TP immunoassays; 5) conducted clinical/immunological evaluations including PET scanning and serum cytokine analyses; and 6) evaluated epidemiologic factors and created novel models to predict syphilis incidence.

Now as one of the few research groups in the world studying syphilis in humans, we propose to further our research program to study the human host immune response to TP, focusing on selected aspects of syphilis immunology, TP biology, and diagnostics, through the following three Specific Aims:

Aim 1: Clinical epidemiology – The study will recruit, treat and follow 150 individuals with incident syphilis without prior infection (recently TP seronegative) and 200 individuals with repeat syphilis infection (recently TP seropositive). We will compare markers of immunobiologic response over time in those 2 cohorts, including RPR and TPPA titers, prototypical inflammatory serum cytokines, and immunologic epitope serum TP antibody assays accounting for HIV-infection status, CD4 T-cell count and HIV viral load. We will also recruit 50 individuals without syphilis as negative controls for immunological assays. All individuals will be followed prospectively with sample collection at months 1, 3, 6, 9, and 12.

Aim 2: Biological — Hypothesizing that the clinical manifestations and immunologic responses will differ between individuals with repeat infection versus *de novo* incident syphilis infection, active versus treated infection, HIV-associated immunosuppression and TP strain type, we will investigate whether a relationship exists during early syphilis between differential gene expression in TP, development of the immune response to treponemal antigens, host cytokine profiles of disease pathogenesis and immune correlates of infection.

Aim 3: Rapid test development/evaluation — Using our current biobank and new specimens from study participants from Aim 1, we will evaluate new rapid tests.

Aim 4: Analyze the host global patient transcriptome to identify diagnostic signatures for syphilis. To find assays that can complement serologic testing for diagnosis of syphilis and to help differentiate between active and resolved infection, we will employ whole blood RNA-seq data from pre-and post-therapy syphilis patients with the overall goal to identify transcription pathways and signatures generated in response to infection. Building on our years of ongoing collaborative work in syphilis pathogeneses and diagnostics, this proposal will focus on new areas of syphilis immunology to inform vaccine discovery and novel diagnostic test development.

Aim 5: Characterize the humoral and cellular immune response in infected patients with and without syphilis history. **Aim 5a**: Identify reactive treponemal antigens associated with infection and reinfection that can serve as novel diagnostic markers and markers of partial immunity. Baseline serum samples will be analyzed by stage of infection using a novel large Treponema pallidum (TP) pan-proteomic array developed by Antigen Discovery Inc. (ADI) to identify immune reactive antigens associated with infection and repeat infection. Because previous syphilis confers partial protection in subsequent infections, we hypothesize that samples from individuals with and without syphilis history will have unique serological patterns that can inform which antigens to use for future vaccine development. **Aim 5b**: Identify TP proteins that activate a CD4 T cell response and elicit Th1/Th2 response during syphilis. Results from Aim 5a will be used to select immunogenic proteins to stimulate peripheral blood mononuclear cells from syphilis patients in vitro to characterize the T cell CD4 immune response by ELISpot and flow cytometry. We hypothesize that individuals with prior syphilis develop T cell CD4 memory response specific to TP antigens and will differ from those without prior syphilis. **Aim 5c:** Analyze T cell receptor (TCR) profile expression during syphilis and post-treatment to explore differences over time and between individuals by status of prior infection. Advanced immunoinfomatics will be used to identify TP-associated TCRs using novel single-cell TCR sequencing developed by Omniscope, linking TCR results to existing databases. We hypothesize differences in TCR expression by treatment and history of syphilis status.

Team roles & responsibilities

As the study PI, Dr Klausner will have the general oversight on the protocol implementation procedures. He will work on protocol development, study design and procedures. Additionally, the USC team will receive coded data and assist in analysis, result interpretation and authoring of papers or presentations about the study.

The UPCH team will be responsible for recruitment, enrolment, specimen, data collection, data management. UCPH-affiliated research associates will be trained on the study procedures and protocol and will be responsible to conduct the screening and informed consent. The UPCH PI will have the on-site overview of the study and be responsible for implementing all study procedures as per protocol.

Methodology

Recruitment, enrollment, specimen collection and testing

The study will recruit 410 individuals with syphilis infection: 150 with *de novo* infection, 200 with repeat infection, 60 with a syphilis lesion, and 50 individuals without syphilis. All individuals with and without syphilis will be followed at 1 month and then quarterly for 12 months. Recruitment will take place in five STI clinics in Lima, Peru, under the guidance of UPCH research personnel: Centro de Salud Alberto Barton (Callao), Centro de Salud Sexual Epicentro (Barranco), Centro de Salud Tahuantinsuyo (Independencia), Centro de Salud Caja de Agua (San Juan de Lurigancho), Centro de Salud Gustavo Lanatta (Chorillos).

At each study clinic, the study team will approach patients newly diagnosed with syphilis and individuals without the infection. Patients will be identified by the clinic appointments, as these cases returning for treatment. A study associate will approach these patients when they enter the exam room for their clinic visit. The study associate will inform the patients of the study and request permission to screen patients for eligibility using the inclusion/exclusion criteria. Interested participants will be informed about the risks, benefits and alternatives of joining the study.

Participants will be screened through (a) a brief medical history/behavioral interview checking for inclusion criteria related to medical/personal history; and (b) rapid tests, including the rapid Abbott Determine™ TP and HIV tests to assess syphilis and HIV infection, respectively. These steps are part of the routine care.

Based on the rapid screening interview and the rapid test results, individuals meeting inclusion criteria will be invited to participate in the study. The research assistant will obtain informed consent.

The **inclusion/exclusion** criteria is as follows:

For Aims 1 and 2, Eligibility criteria include:
1. Age > 18 years
2. Newly diagnosed with early syphilis infection
   a) Group 1 (n=150): No history of syphilis and documented negative T. pallidum antibody test in past 12 months.
   b) Group 2 (n=200): History of syphilis (documented positive T. pallidum antibody test) and history of syphilis treatment
   c) Group 3 (n=60): No documented history of syphilis. Presence of a lesion compatible with early syphilis.
   d) Group 4 (n=50): No history of syphilis, documented negative T. pallidum antibody test in past 12 months and negative syphilis tests at enrollment visit.
3. Intention to stay within Lima or its environs for the next 12 months

4. Willingness to provide the required study samples (venipuncture whole blood specimens and swabs from genital lesions)

Exclusion Criteria:
1. Unable to give informed consent
2. Unlikely to complete study follow-up
3. Signs or symptoms indicative of neurosyphilis

Case definition:

New (de novo) syphilis infection: Individual with a newly reactive RPR, TPPA-confirmed positive, who has a documented prior TP negative antibody test taken within the past 12 months at the STI facility. Existing health records must include the syphilis testing history.

Repeat syphilis infection: Individual with a new 4-fold rise or newly reactive RPR when previously RPR seronegative, with a documented prior TP antibody positive test at the STI facility.

All enrolled participants will receive standard of care HIV/STD pre-test counseling, after which a trained laboratory technician will collect a blood specimen for HIV rapid testing, and venipuncture whole blood for syphilis testing (RPR, TP antibodies) and the necessary blood samples for the study testing (Proteomics/microarray; PBMCs for ELISpot, flow cytometry and TCR repertoire immunoassays, and RNA-seq) as well as lesion swabs for TP isolation, when lesions are present. Additionally, if lesions are present, non-identifying photographs of the lesions will be requested to document clinical characteristics and stage as well as to aid clinical confirmation.

We will collect contact information from participants to use to help with follow-up and retention. With the participants' consent, we will text or call them prior to their scheduled follow-up visit as a reminder. Should they fail to present for the visit we will make up to 7 additional attempts via text, telephone or social media (e.g, Facebook) to contact them.

Recruitment will continue until each group reaches their enrolment targets.

Study visits: Baseline, 4, 12, 24, 36 and 48 weeks

Participants from all 4 Groups will undergo 48 weeks of follow-up after enrollment and will be retained through our previously validated successful methods of weekly text messages, follow-up calls and notifications. If individuals are re-infected with syphilis during their follow-up, they will be asked to attend a new visit at 4 weeks and then continue with their quarterly visits.

- At baseline, testing will include RPR, TP antibody testing, HIV testing, immunoassay serum antibody testing, whole blood for PBMC studies, plasma for proteomic/microarray and swabbing from lesions, if present, for TP DNA detection, strain characterization and storage . Additionally, baseline data will be collected using a survey to collect sociodemographic variables, as well as data on PrEP use, and HIV care experience. All samples should be collected prior to syphilis treatment administration for syphilis cases.
- At 4 weeks, the visit will include RPR testing and repeat whole blood for RNA-seq and PBMC studies, serum and plasma specimen collection for proteomics/microarray.

- For groups 1, 2 and 3: At 12, 24,36 and 48 weeks RPR testing, repeat HIV testing among those HIV-negative at baseline; whole blood for RNA-seq and PBMC studies, serum and plasma specimen collection for proteomics/microarray testing will be conducted when RPR titer declines as indicative of cure.

All cases will be treated for *Treponema pallidum* infection per Peru guidelines and cases of HIV infection will be linked to appropriate services as per clinic protocol. HIV and syphilis results will be given to participants using appropriate post-test counseling. Per Peru and CDC guidelines, syphilis treatment will be provided—benzathine 2.4 MU intramuscular injection once for primary, secondary or early syphilis. TPPA confirmatory test results will be given to study participants 7-10 days after the initial visit. Study-related laboratory results will not be disclosed to participants.

Partner treatment and referral. After a new HIV or syphilis diagnosis, counselors ask participants to inform and bring his partner(s) for evaluation. If that is not feasible, counselors encourage the participant to refer partner(s) to a Ministry of Health clinic or other health center for syphilis treatment and/or HIV testing. As per Peruvian standards of care, if the patient is diagnosed with syphilis and declines to bring in or refer partners, we will offer expedited partner therapy: doxycycline 100mg tablets 2/day for 14 days. Counselors will also navigate linkage for HIV care or any other referrals (e.g., medical, social, and/or mental health), promote partner treatment uptake and suggest partners attend to health care centers to receive care in person.

Study laboratory testing and procedures

Specimen handling and sample storage. Whole blood (35.5ml) samples will be collected for PBMC isolation and RNAseq (for Aims 2b and 3) and for plasma and serum extraction. Specimens will be initially tested at each clinic site following standardized laboratory protocols. Plasma and serum will be separated at each site's laboratory and then transported to the UPCH Sexual Health Laboratory along with whole blood. Each sample type will be stored appropriately for the downstream studies. Once at the Sexual Health Laboratory, TPPA testing will be conducted (SERODIA® TP•PA, Fujirebio Inc., Japan). Samples will be stored at -80ºC for up to 10 years after the end of the study for additional testing and future studies, if approved by the participant during the informed consent process. PBMC isolation involves separating blood components by density-gradient centrifugation using Ficoll for separation. This procedure will be performed at the Sexual Health Lab, as we have done in the current ongoing study. Briefly, PBMC will be isolated by density gradient centrifugation using Ficoll, then cells will be counted and cryopreserved, using standardized protocols. For RNAseq, whole blood will be collected in Tempus Blood RNA tubes for stabilization and isolation of total RNA for gene expression analysis. Advantages of Tempus Blood RNA tubes include immediate lysis of blood cells and RNA stabilization in a single step, with no RBC lysis or proteinase K treatments required. Samples will be stored at -80ºC while awaiting extraction, DNase treatment, and library preparation prior to sequencing. To collect lesion samples, staff will don nitrile gloves. After cleaning the lesion surface with antibiotic-free sterile saline solution, the lesion exudate will be collected using a synthetic swab (Puritan Brand) and placed in a vial with 500µl of DNA extraction buffer. Once the samples have been collected, they will be stored at -20°C until they are sent to the Sexual Health Laboratory at UPCH, maintaining the cold chain, where they will be stored at -20°C until further processing. TP DNA will be extracted and tested using specific primers in conventional polymerase chain reaction (PCR) to amplify TP0547 and TP0548 target genes and the human β-globin gene as an internal control. We have successfully implemented those methods in prior studies

Syphilis testing: According to Peru's guidelines, syphilis testing is offered to all individuals seeking STI services. Study screening will use onsite RPR and rapid point-of-care syphilis TP antibody tests (Alere Determine™ Syphilis). Reactive RPR tests will be quantified to the maximal reactive titer. At the UPCH Sexual Health Lab, serum TPPA testing will be performed (reactive cutoff value of ≥1:80). Four additional EDTA-containing tubes of whole blood will be collected for subsequent testing including: TP antigen testing and selection, PBMC recovery for ELISpot, flow cytometry assays, and TCR repertoire analysis, and for transcriptome assays.

HIV Testing: After consent, all participants will undergo rapid HIV-1/2 antibody testing on serum using two 4th generation HIV-1/2 antibody/antigen immunochromatographic tests (Determine HIV early detect, Abbott, USA; OnSite HIV Ag/Ab 4th Gen, CTKBiotech, USA). We do expect that at least a third of potential participants to be living with HIV. For individuals living with HIV, we will ask for permission to retrieve their CD4 T-cell counts and HIV viral load from their clinical records.

Molecular syphilis testing: Complementary testing of genital lesion exudates will be performed on Tp47 target positive samples following the enhanced CDC genotypic sub-typing protocol as we have previously described.

Cytokine testing: Given the success of our prior work describing unique cytokine patterns in patients with early syphilis, we will collect serum samples from participants at baseline and at each follow-up visit after the date of syphilis diagnosis. Whole blood will be drawn into silicon-coated Vacutainer tubes, separated by centrifugation, and serum frozen at -80°C. Specimens will be transported on dry ice to undergo testing for 62 cytokines by Dr. Holden Maecker at the Stanford University Immunology Core Laboratory (see Letter of Support). A 62-plex cytokine assay system will be used to determine a panel of cytokine responses including the known Th1/Th2 cytokines IL-2, IL-4, IL-6, IL-8, IL-10, IL-17, IFN-gamma, and TNF-alpha (Luminex multiplex assay, Millipore, LINCO Research Inc, St. Charles, MO). All measurements will be performed in duplicate, masked from clinical data. Values <2 pg/mL will be assigned 2 pg/mL.

Elispot and flow cytometry to identify cytokines.

Immunoassay array of immune response to syphilis infection: Our team has previously expressed and characterized the immune response to several TP lipoproteins and surface-exposed proteins. For this proposal we will use our research-developed ELISA to investigate the humoral response to a panel array of TP lipoproteins (LPs), ligand binding proteins (LBPs) and putative surface-exposed proteins (SEPs), selected based on their immunogenicity, and potential for differential expression in different stages of early infection (primary, secondary and early latent).

Immunoinfomatics to identify TP-associated TCRs.

Transcription pathways for syphilis diagnosis.

Study Outcomes

Study Aim 1: Compare markers of immunobiologic response to de novo versus repeat infection including RPR serologic patterns and titers, cytokine profiles and serial immunologic epitope assays accounting for HIV-infection serostatus, CD4 T-cell count and HIV viral load. The goal of this aim is to evaluate the changes in humoral immune response against selected TP antigens in patients with a new or repeat infection. Such antigens include TP LPs, LBPs, and SEPs. Approach: Antigens will be expressed using recombinant DNA technology as already done for TP0435,73 TP0126,74 TP0136,75 TP0620,72 TP0621,72 and TP089772 in the laboratories of Dr. Giacani at UW or as already described by other investigators for the remaining antigens.90,99-103 Briefly, recombinant antigens will be expressed in E. coli and purified by nickel-affinity and size-exclusion chromatography. Proteins that cannot be expressed as soluble antigens will be purified under denaturing conditions using 8 M urea or 6 M guanidine HCl, and dialyzed against a buffer containing proteins stabilizers (e.g. arginine HCl) and detergents (e.g. OTG, BriJ-35, or glycerol) to avoid protein precipitation. Preparations with sufficient purity will be used for antibody detection in ELISA format as already described.72 Levels of target-specific 3Tantibodies33T will be compared for the same stage as well as between 33Tdifferent33T 33Tstages33T of syphilis, and in new vs repeated infections.

Expected results: Based on previous work,54,98 in new syphilis cases we expect to see reactivity against TP0435 to be similar in primary and secondary syphilis and decline in early latent cases. Reactivity to TP0574, TP0769, and TP0136 is expected to be highest during primary syphilis and quickly decline in subsequent stages, while reactivity against TP0768 and TP0684 should be similar in primary and early latent syphilis but lower in secondary syphilis sera. Reactivity against TP0163 and TP1038 should be highest in early latent cases, compared to primary or secondary cases. Response to the other antigens should fall within one of the above groups. We will also determine if the humoral responses detected here correlate the message RNA level for most genes that will be detected in Aim 2.

Study Aim 2: To investigate whether a relationship exists during early syphilis between differential gene expression in TP, development of the immune response to treponemal antigens, and host cytokine profiles to identify molecular markers of disease pathogenesis and novel vaccine candidates.

Methods. To obtain treponemal samples for total mRNA-sequencing, swabs of syphilitic lesions on the mucosal tissue or skin will be obtained from enrolled patients presenting to the STI clinic with primary and secondary syphilis lesions (i.e. a primary chancre, disseminated skin lesions, or condylomata lata, where treponemes are known to be highly abundant). Collected samples will be immediately frozen in liquid nitrogen using a cryo-preservative solution such as the one used to preserve viable stocks of TP isolates (i.e. a saline solution supplemented with 20% glycerol and normal rabbit serum) and be shipped to our laboratory at UW. Prior to total RNA extraction, treponemes will be separated from the host debris by incubation with streptavidin-coupled magnetic beads coated with a biotinylated anti-TP serum (obtained from long-term infected rabbits).

Following treponemal enrichment, total RNA will be extracted from the sample using a ChargeSwitch Nucleic Acid Purification Kit (Invitrogen), which includes a DNase incubation step to eliminate residual genomic DNA. This procedure will be followed by mRNA enrichment using the MicrobExpress and MicrobEnrich kits (Life Technologies), reverse transcription, total RNA amplification using a SeqPlex RNA Amplification Kit (Sigma) and library preparation. Expected results and interpretation. We anticipate that comparative analysis of TP transcriptomes will reveal differences in transcription of virulence-associated genes, particularly those encoding for TP LPs, known to be the main antigens recognized by the host TLR receptors during syphilis infection (as mentioned previously, TP lacks LPS). In particular, we expect that LP expression will positively correlate with the level of TLR2-mediated proinflammatory cytokines such as TNF-α, IL-1β, IL-6, and IL-12. We also expect to see modulation on transcription of genes controlled by AσF. Those include several of the pathogen's motility and chemotaxis genes, and genes that are involved in detoxification of reactive oxygen species (ROS) also shown by our group to be controlled by an AσF.106 It is known that during primary experimental syphilis, treponemes are cleared from the primary lesion by opsono-phagocytosis following the appearance of anti-TP specific antibodies.124 Despite this, a sub-population of treponemes resistant to macrophage ingestion emerges and persists,125 suggesting that treponemes modify the availability of surface proteins targeted by opsonic antibodies. Based on this evidence, we expect to find a decrease in transcription of several (but not all) genes encoding TP putative OMPs in samples harvested from patients with high titers of treponemal antibodies. Such differential expression could explain the increased resistance to opsono-phagocytosis of the syphilis agent. As mentioned, however, we do not expect to see down-regulation of all TP putative OMPs. We recently reported that transcription of TprK, a putative OMP, does not change throughout the course of primary experimental syphilis.91 This is likely possible because the TprK protein escapes immune detection via antigenic variation of its surface exposed regions,104,126,127 and its expression does not necessarily need to be downregulated during infection. We also expect to see variability in expression of additional putative OMPs with proteolytic activity, and ability to bind the host extracellular matrix components such as Tp0751128 or Tp0136,129 amply studied by our group, which are likely involved in tissue invasion and dissemination. Too little is known about how post-transcriptional mechanisms affect TP proteome, even though work by McGill et al.54 strongly suggests that protein expression in 17TTP17T is consistent with mRNA levels.

AIM 3: Rapid Point-of-Care test development/evaluation – We currently have a biobank (n ≥ 3000 serological and clinical specimens) which, along with new specimens from Aim 1, will be used towards the development and evaluation of additional diagnostic tests for syphilis infection. We will test the biobank samples on new and under development rapid tests to evaluate their performance.

Aim 4: **Analyze the host global patient transcriptome to identify diagnostic signatures for syphilis.** As described in Aim 1, we will appropriately collect blood from syphilis patients to preserve RNA for gene expression studies. Initial trials, however, will be conducted using previously collected de-identified clinical and preclinical samples unrelated to this study to optimize the sequencing pipeline. Given the enrollment estimates, we expect to receive sufficient samples to initiate the planned work in the second quarter of year 2. Total RNA will be extracted from samples collected in Tempus tubes using Tempus Spin technology (Applied Biosystems) and depleted of alpha-globin and beta-globin mRNAs using globinCLEAR protocol (Ambion) using 1 μg of total RNA as input. Subsequently, 400ng of globin-depleted total RNA will be used for library synthesis with TruSeq Stranded mRNA HT kit (Illumina). Libraries will be quantified using qualitative PCR with PerfeCTa NGS kit (Quanta Biosciences) or equivalent method. Equimolar amounts of samples will be pooled and clustered on a high output flow cell using HiSeq SR Cluster kit V.4 (Illumina). Approximately 20-30 million paired end reads per sample will be collected and pseudo-aligned to the human reference transcriptome. RNA-sequencing will be performed at Dr. Alex Greninger's (Co-I) UW lab. Dr. Greninger has recently been collaborating with our group on TP genome sequencing.112 The raw sequencing data will be preprocessed using bcl2fastq software and the quality assessed using FastQC tools. Differential expression analysis and a custom bioinformatics pipeline will be used for unbiased detection of molecular signatures of specific cell types associated with the host response to TP infection, reinfection, and clearance. De-identified, processed RNAseq counts data and sample metadata will be publicly available in the NCBI GEO database. Importantly, immune responses and T cell immune polarization observed in human samples identified in this Aim will complement the T cell studies proposed in Aim 2b. T cell immune polarization refers to their ability to produce either T1 or T2 patterns of cytokines: interleukin (IL)-12 and interferon γ on the one hand (T1), and IL-4 on the other (T2). The T1 and T2 T-cell subsets mediate largely non-overlapping immune functions that are specialized to deal effectively with different classes of pathogens. Through RNAseq, polarization will be ascertained by detecting specific mRNA signatures that are associated to the T1 (IL-2, IFN-gamma, TNF-alpha/beta) or T2 (IL-4/5/6/10) pathways.

Expected results. Upon performing RNA-seq on all samples, we expect that principal-component analysis will clearly separate the samples collected from the healthy subjects from the acutely infected, and the early and late convalescent ones. Significant differentially expressed transcripts (DETs) will be defined as those having a P value ≤ 0.05 and at least a 2-fold change in expression at any time point relative to uninfected subjects. We expect to identify several hundred DETs, with the DETs number gradually increasing in samples from patients where the disease has progressed toward secondary stages. DETs should then gradually decrease as manifestations resolve naturally due to the host-mediated clearance of the pathogen, during the latent phase or as the result of treatment. We expect however to still detect differences in samples collected from subjects with latent (untreated) infection compared to samples collected post-treatment, with the latter virtually undistinguishable from healthy subjects. As for temporal gene expression changes, we expect samples from the healthy donors to display a relatively broad range in intensity values, likely reflecting normal variation in gene expression in the population. We also expect that the range of normalized intensities will appear to be more restricted in the acute syphilis samples when compared to samples from the healthy controls, likely reflecting a common response to infection with TP antibodies. Due to the Th1 nature of the immune response that develops to treponemal infections, we expect to see the greatest expression changes in interferon (IFN)-regulated genes. Overall, we anticipate that our approach will provide fundamental data to help understand the host reaction to infection and the signatures associated to development of partially protective immunity. These data might also help design improved diagnostics for syphilis. We will generate a set of gene classifiers that can accurately separate subjects with syphilis from healthy controls, and patients that are actively infected at their first episode vs. those that are re-infected after initial treatment. This approach will have a critical advantage over current serologic tests offering the potential for more accurate diagnosis of early syphilis and may also allow improved monitoring of treatment efficacy and disease resolution.

Aim 5: **Characterize the humoral and cellular immune response in infected patients with and without syphilis history**. **Aim 5a: Identify reactive treponemal antigens associated with infection and reinfection that can serve as novel diagnostic markers and markers of partial immunity** Patients with a history of treated syphilis have a unique memory-driven immune response compared with those with a first time infection.119 One manifestation is an attenuation of early clinical manifestations and an absence or shorter duration clinical lesions.20 That response, however, is unable to generate complete immune protection against infection, i.e. not “sterilizing immunity”. Better defining the memory-driven immune response could point at several antigens that could be used for future vaccine development to prevent clinical disease (i.e., chancres, sores, lesions, rashes) and because clinical disease is what drives the spread of infection, subsequent transmission.27,120 Here, samples from individuals with and without syphilis history will be analyzed using a novel TP pan-proteomic array to identify responsive antigens that can be tested as future vaccine candidates. Furthermore, analysis of sera pre- and post-treatment could lead to the identification of antigens that could serve as novel markers of active infection.

Fabrication of TP proteome array. A library of 1,043 expressible TP open reading frames (ORFs) has been established at Antigen Discovery, Inc. (Irvine, CA). Each protein, epitope, domain, or protein fragment’s gene sequence was amplified by PCR and inserted into the vector pXT7 by recombination in E. coli to establish a library of partial or complete coding DNA sequences. Each clone has been quality-checked by two methods: 1) staining of PCR products after gel electrophoresis to assess successful amplification and correct size, and 2) sanger sequencing verification. Among those 1043 ORFs are 16 genes unique to the TP SS14 strain and 15 genes unique to the TP Nichols strain-all remaining core genes were cloned from the Nichols strain. Preliminary data will guide the selection of antigens for inclusion in the proposed study using a cost-effective, down-selected chip (preliminary data from SBIR R43AI149804, are currently being analyzed). Up to 200 of the full proteome library will be selected and expressed using a coupled E. coli cell-free in vitro transcription and translation (IVTT) system and spotted onto 16-pad nitrocellulose-coated glass AVID slides (16 array replicates per slide, up to 200 antigens per array) using an ArrayJet Marathon Argus robotic non-contact microarray printer. Each protein array will contain controls for systematic and biological effects. Human immunoglobulins (IgG, IgA will be spotted on each subarray at multiple dilutions to control for secondary antibody reactivity. Additionally, anti-human immunoglobulin monoclonal antibodies will be spotted to capture IgG and IgA in samples as a probing control. At least 16 IVTT spots without target DNA will be included as controls for biological cross-reactivity and normalization. Each expressed protein will include a 5’ polyhistidine (His) epitope and 3’ hemagglutinin (HA) epitope. Array chip printing and protein expression will be quality checked by probing random slides with anti-His and anti-HA monoclonal antibodies fluorescently labeled and quantifying spot signals using a microarray scanner. Purified recombinant proteins, some commonly used in recombinant treponemal tests such as EIA and CIA, will be printed on the arrays for comparison of IVTT reactivity with purified protein reactivity.

Protein microarray sample probing, data extraction and pre-processing. Plasma samples will be diluted (1:100) and incubated on the arrays overnight at 4ºC. Bound IgG will be detected with Cy5-conjugated anti-human IgG Fcɣ that can be detected by the “red” laser filters. Bound secreted IgA will be detected with Cy3-conjugated anti-human IgA that can be detected by the “green” laser filters. Chips will be scanned and quantified using a GenePix microarray scanner and GenePix Pro software. Foreground spot intensities will be adjusted by local background by subtraction and negative values will be converted to 1. Next, all foreground values will be transformed using the base 2 logarithm. The dataset will be normalized to remove systematic effects by subtracting the median signal intensity of the IVTT controls for each sample. Since the IVTT control spots carry the chip, sample and batch-level systematic effects, but also antibody background reactivity to the IVTT system, this procedure normalizes the data and provides a relative measure of the specific antibody binding to the non-specific antibody binding to the IVTT controls (a.k.a. background).

Expected Results. We anticipate that about 40% of the *TP* proteome will be recognized, and that there will be overlap between reactive antigens in patients with and without syphilis history. However, we expect more elevated antibody responses to key antigens in those with a history of past infection(s), particularly to surface antigens that might represent targets of opsonic antibodies and mediate pathogen clearance. These antigens could represent vaccine candidates to be tested in future pre-clinical immunization/challenge experiments. IgG and IgA against *TP* proteins in our subjects will help define antigens that elicit antibodies rapidly and either are maintained over time or decline in response to treatment. These data could lead to the identification of staging markers. From the analysis of pre- and post-therapy sera, we expect to see a decrease in reactivity to several antigens after treatment, which could be considered in lieu of non-treponemal tests to assess successful therapy.

Aim 5b: Identify *TP* proteins that activate a CD4 T cell response and elicit a Th1/Th2 response during syphilis. *In silico* predictions of CD4 T cell epitopes in *TP* proteins will be performed to select those most appropriate for synthesis. Using ELISpot and flow cytometry (FACS), PBMCs from syphilis patients will be analyzed for CD4 T cell cytokine production and activation markers after recombinant *TP* antigens stimulation to identify potential immune targets. Our primary hypothesis is that memory T cell responses develop in individuals with prior syphilis, and this immune response is associated with unique clinical outcomes. From the array of proteins identified in Aim 5a, *in silico* analysis will be performed to predict the helper T cell epitopes in the *TP* proteins using: (tools.iedb.org/mhcii). The immunogenicity of the selected epitopes will be evaluated using (tools.iedb.org/CD4episcore). Then, high scoring epitopes will be analyzed using the IFNepitope tool (crdd.osdd.net/raghava/ifnepitope) to predict the ability to induce IFN-γ production[122,123]. *TP* proteins covered with CD4+ T cell epitopes and with a high score for the evaluated parameters will be selected and synthesized for *in vitro* testing using ELISpot (IFN-γ production) and flow cytometry (CD4 T cells expressing CD69, IL-2, TNF-α, IL-4, and IFN-γ) using PBMCs from individuals diagnosed with syphilis. ADI will produce these proteins using a cell-free E. coli expression system.

ELISpot assays for IFN-γ: ELISpot plates will be treated/activated with 70% ethanol and washed three times with PBS. Then, wells will be coated with 100ul of IFN-γ capture antibody and stored overnight at 4°C. After wash, wells will be blocked using 100ul of bovine serum albumin solution for 2h at 37°C. Fifty thousand PBMC/well will be added in Roswell Park Memorial Institute (RPMI) -10% fetal bovine serum and stimulated with the previously identified/selected *TP* recombinant proteins or medium alone for 24h at 37°C in 5% CO_2. A positive control will be included in all assays: mitogens e.g., PHA/ionomycin. After incubation, cells will be discarded and wells washed with PBS containing 0.1% Tween-20 and then coated with 100ul of biotinylated anti human IFN-g in PBS-Tween. After 3 hours of incubation plates will be washed and 100ul of avidin-peroxidase-complex added at the appropriate dilution.[124] Following incubation for 1 hour at room temperature, plates will be washed with PBS-Tween and then with PBS finally adding acetic acid-3amino-9ethylcarbazole and stopping the reaction with distilled water. The results will be expressed as the frequency of IFN-g producing cells in PBMC.[124]

***TP* specific CD4 T cell stimulation:** This assay will stimulate PBMCs, as a source of CD4+ T cells, with the *TP* proteins previously selected *in silico* and using ELISpot. Controls include a non-stimulated negative control and a leukocyte activation cocktail (BD Biosciences)-stimulated positive control. During stimulation a protein transport inhibitor (monensin; BD Biosciences) will be added to all cultures. After 16 hours of incubation at 37°C, cells will be washed with PBS-1% FBS and surface stained with fluorescent antibodies: anti-CD3, anti-CD4, anti-CD8, and anti-CD69. After incubation for 20 minutes at room temperature in the dark, intracellular staining will be performed using 300µl Cytofix/Cytoperm solution (BD Biosciences), incubated for 15 minutes at 4°C in the dark, then washed twice with BD Perm/Wash buffer (BD Biosciences), then stained with fluorescent antibodies anti-IFN-γ, TNF-a, anti- IL-4 and anti-IL-2 cells and incubated for 25 minutes at 4°C in the dark, then washed twice with BD Perm/Wash buffer, and resuspended in PBS for data acquisition on an FACS CANTO II (BD Biosciences). Gating strategy: $CD3^+CD4^+CD8^-$ T cells will be gated separately and each group interrogated for markers/cytokine-positive cells (CD69, IL-2, TNF-α, IL-4 and IFN-γ) for all treatments: stimulation, positive and negative controls. Flow cytometric analyses will be performed using FlowJo V10 (ThreeStar, Ashland, OR). **Expected Results.** These experiments will indicate which TP proteins are recognized by CD4 T cells and elicit IFN-γ production. We anticipate that the selected proteins based on *in silico* will be able to stimulate IFN- α production by CD4 T cells. These experiments will be important to the design the planned FACS experiments. To identify IFN-γ CD4+ T cells producing, intracellular cytokines staining by FACS will be an appropriate approach to define the Th1/Th2 profile. We predict that results from these experiments will validate our hypothesis that *TP* proteins are recognized by CD4 T cells from individuals with previous infectious and induce significant cytokines production. The results will allow us to determine TP antigens that produce a robust Th1 response associated with protective immunity and that have the potential to become vaccine candidates.

Aim 5c. Analyze TCR profile expression during syphilis and post-treatment to explore differences over time and between individuals: We will send Omniscope 20 paired samples (40 total) over years 1-4 of the grant for this in-depth analysis. Samples would include five individuals with syphilis history and HIV co-infection, five individuals with syphilis history and without HIV co-infection, five individuals without syphilis history with HIV co-infection, and five individuals without syphilis history and without HIV co-infection. Each individual would act as their own control, with one specimen analyzed at diagnosis and another once clinical response to treatment has been documented. The resulting TCR repertoires would be analyzed to document the number of unique clonotypes by sample and time-point (pre and post treatment). We hypothesize that syphilis treatment should reduce the antigen burden and affect T cell clonal expansion and TCR diversity, also individuals with prior infection would have memory T cells with unique profile compared to individuals without prior infection. However, these analyses will be hypothesis generating rather than hypothesis driven. The analyses planned in Aim 5c are exploratory only and are not powered to measure a specific effect.

Data management

All study data will be collected and managed by the Center for Interdisciplinary Research in Sexuality, AIDS, and Society (CIISSS) at UPCH, directed by Dr Cáceres (PI). Analyses will be conducted by Dr. Konda (Co-investigator; epidemiologist) at USC. Dr Konda is currently based in Peru and will conduct frequent site visits with the study team at CIISSS in the first year of the study to ensure that data sharing protocols are in place. CIISSS has substantial experience in data collection and management for NIH-funded trials and industry-sponsored studies. The analytic database will only contain a single unique study identifier for each patient, and will not contain any other identifying individual information. The laboratory results will only be registered with sample ID and linked to the results from the tests. Access to the information will be controlled with user IDs, secure password protection, and multi-level rights that grant different privileges to members of the research team. Additionally, all sites will use barcode readers to log all laboratory specimens collected using the unique participant study ID. All user connections to the server will be encrypted using industry-standard 256-bit SSL encryption. All clinical and lab data collected on study participants for Aims 1 and 2 will be done through REDCap (https://www.project-redcap.org/), a web-based and HIPAA-compliant data collection system that is currently used by both CIISSS and USC. Data is accessed through a password-protected web portal, making it secure and easy to share data between institutions in different countries. Existing biobank data (Aim 3) will be shared with the USC research team through this same portal.

Data Analysis

General Aspects. Comprehensive descriptive and exploratory analyses will be conducted with information collected before testing the hypotheses of the study. Frequency distributions will be tabulated for all variables, performing range checks and cross-validations, and evaluating individually all data points with measurements 3 standard deviations above their mean. Missing or erroneous data will be coded as such and frequencies of unusable data will be determined for every variable, determining if the frequency was unusually high and affected the validity of the data. *Aim-specific Analyses.* Aim1. Primary analyses will be conducted for Aim 1 and will use random-effects (RE) logistic regression models to evaluate the impact of *de novo* versus repeat infection on a binary outcome for the probability of asymptomatic vs. symptomatic or complicated disease (based on the clinical evaluation by group). RE will be included to account for correlations between repeated outcome measurements on the same study participants. Given the frequent presence of HIV co-infection (20-40%), HIV-infection will be included as a key three-category confounder in all analyses: individuals without HIV-infection, those with controlled HIV-infection (i.e. with undetectable viral load), and those with uncontrolled HIV-infection (i.e. those with detectable viral load > 200 copies/ml). All analyses will also include covariates to account for linear or nonlinear trends in outcomes over time, based on formal statistical tests and visual inspection of plots that show outcome trajectories over time. RE regression will also be used to evaluate the impact of *de novo* versus repeat infection on secondary outcomes, including: 1) epitope immunoassay responses, 2) RPR pattern (as RPRs are non-linear, this will be modeled using log-poisson GEE), and 3) cytokine profiles by group.

Appropriate regressions will be used for the type of outcome modeled. For example, RPR titers will be classified as non-reactive, 1:1, 1:2, and so forth in an ordered fashion; RE ordinal regression will be used. Exploratory analysis will be focused on individuals who are serofast, defined as individuals who persist at RPRs of 1:4, 1:2, or 1:1 for more than 12 months, who will be identified within the existing biobank from the previous cohort and compared to those to individuals from the current cohort with known active (repeat or *de novo*) infections of the same titer. Aim 2. We expect to find different response profiles for each normalized TP gene expression over time. Additional covariates will be added to RE regression models from Aim 1 to compare clinical manifestations and immunologic responses by TP strain type, as well as by de novo / repeat infection and HIV infection status. Aim 3. Sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) will be calculated, along with 95% confidence intervals, to evaluate the performance of each new rapid dual HIV / syphilis test. We will further stratify the TP antibody results by RPR titer of the specimens to assess the sensitivity at the limit of detection. McNemar's test will be used to compare sensitivity, specificity, PPV, and NPV between each new test and a standard syphilis test. Aim 4. Upon performing RNA-seq on all samples, we expect that principal-component analysis will clearly separate the samples collected from the healthy subjects from the acutely infected, and the early and late convalescent ones. Significant differentially expressed transcripts (DETs) will be defined as those having a P value ≤0.05 and at least a 2-fold change in expression at any time point relative to uninfected subjects. We expect to identify several hundred DETs, with the DETs number gradually increasing in samples from patients where the disease has progressed toward secondary stages. DETs should then gradually decrease as manifestations resolve naturally due to the host-mediated clearance of the pathogen, during the latent phase or as the result of treatment. We expect however to still detect differences in samples collected from subjects with latent (untreated) infection compared to samples collected post-treatment, with the latter virtually undistinguishable from healthy subjects. As for temporal gene expression changes, we expect samples from the healthy donors to display a relatively broad range in intensity values, likely reflecting normal variation in gene expression in the population. We also expect that the range of normalized intensities will appear to be more restricted in the acute syphilis samples when compared to samples from the healthy controls, likely reflecting a common response to infection with TP antibodies. Due to the Th1 nature of the immune response that develops to treponemal infections, we expect to see the greatest expression changes in interferon (IFN)-regulated genes. Overall, we anticipate that our approach will provide fundamental data to help understand the host reaction to infection and the signatures associated to development of partially protective immunity. Aim 5. Aim 5a. Paired T-tests and linear mixed effects regression models will be used for comparison of longitudinally sampled plasma. T-tests (or Wilcoxon tests) and multivariable linear regression models will be used for comparisons between groups at individual time points. Where appropriate, regularization for variance shrinkage will be applied to statistical tests (e.g., empirical Bayes moderated T-tests). All statistical tests will be two-sided. A global alpha level of 0.05 will be used to assess statistical significance, and significance levels will be adjusted for the false discovery rate using the Benjamini-Hochberg method. Generalizability will be assessed using leave-one-out cross-validation (LOOCV). Ability of antibody responses to discriminate subjects based on disease stages will be tested by calculating the area under the receiver operator characteristics (ROC) curve (AUC). Combinations of biomarkers will be tested for discriminatory capacity by feature selection using lasso and elastic net penalization, implemented in the glmnet package in R. Further supervised may be done using PLS-DA, and unsupervised methods may be used on the dataset, such as t-stochastic neighbor embedding (tSNE). Aim 5b, Our primary hypothesis is that memory T cell responses develop in individuals with prior syphilis and this immune response is associated with clinical outcomes. ELISpot: To define the specificity (positivity) of the T cell response to antigens it should meet the following criteria: stimulation index ≥2, at least 100 SFC/10^6 PBMC and p≤0.05 (student T-test) comparing the three groups of individuals those with prior syphilis history, those without prior syphilis history, and healthy controls. Significant group differences will first be determined by Wilcoxon-Mann-Whitney test. Then Student T-tests will be conducted to compare each pair, with the primary comparison of interest being those with and without prior syphilis history. Flow cytometry: Wilcoxon-Mann-Whitney and Wilcoxon rank sum tests will be used to compare the group differences using the same strategy as in ELISpot to determine the percentage of positive T cells to IFN-γ, TNF-a, IL-2, IL-4 and CD69 markers.

Sample Size Calculations focused on binary outcomes (e.g., asymptomatic vs. symptomatic) typically require larger sample sizes to achieve sufficient power relative to continuous outcomes, such as mRNA levels. For the primary analysis in Aim 1, calculations were carried out to achieve at least 80% power at a 0.05 two-sided significance level for a sample size of 90 in each of two groups (total n = 180), assuming 20 of the proposed sample size of 200 (i.e., 10%) will be lost to follow-up. We hypothesize that 80% of those with repeat syphilis (= group 1) will have asymptomatic disease. Based on a range of group 1 proportions from 60% to 90%, we will be able to detect percent differences in asymptomatic disease between those with repeat syphilis and those with *de novo* infection (= group 2) as small as 21% to 16%, respectively. We also note that we will be able to detect fairly small effect sizes (ES) for continuous outcomes; e.g. if we assume 6 repeated measurements (from baseline to 48 weeks) with an autocorrelation of 0.5, we will be able to detect ES as small as 0.32. Calculations were also carried out for Aim 3 to evaluate the performance of dual syphilis / HIV tests for a range of values. A much larger sample size is anticipated as we will combine specimens from our sample of study participants and a biobank; we anticipate a sample of at least 200 to 400 usable specimens, and possibly more.

Table 4. Aim 3 sample size calculations for diagnostic test evaluation

	Sample Size	**Est. Sensitivity**	**95% CI**	**Precision**
Syphilis: Est. prev. = 20%	200	95%	(83.1, 99.4)	16.3
HIV: Est. prev. = 10%	200	95%	(75.1, 99.8)	24.7

Safety Considerations

Possible Risks and Benefits Associated with Study Procedures:

The primary potential risks to participants due to study participation are psychological and social in nature. Physical risks from phlebotomy, fingerprick, oral-pharyngeal swab and lesion swabs for testing as described above are minimal (i.e. mild discomfort, bruising, minor risk of infection) and are no greater than those encountered during routine clinical care.

Psychological: Participants could experience psychological distress such as anxiety when discussing issues related to personal experiences. However, we do not expect any serious events to occur based on our experience across multiple previous studies, including the prior PICASSO cohort, which forms the foundation for this new research. Participants may experience some stress related to the knowledge of STI or HIV status. Participants may experience discomfort with the idea of a photograph being taken of their body. They will be asked to provide informed consent for any photograph taken, only in the presence of a syphilis lesion and all photographs would be non-identifiable, stored only with their study ID, and taken with a clinic camera not a personal camera or cell phone.

Participants will be given information and education about the nature and consequences of syphilis infection and treatment, and those testing positive will be provided treatment as per standard treatment protocols. The likely harmful consequences of learning one's STI status are low.

Social: Participation in this study may cause social harm (e.g., discrimination, false assumptions, rumors) if the participation of the participant becomes known to others.

The risk for participating in this study include physical, social and psychological and they are not greater than those encounter when the patient/participant visits the clinic. Study personnel and clinic personnel, where study activities will occur, are properly trained to handle the above mentioned concerns and offer the necessary support or linkage to proper services.
On the other hand, understanding the pathogenesis of syphilis will allow us to develop novel diagnostic methodologies, vaccines and will affect the care of people infected with syphilis both in Peru, and internationally. We estimate that the benefits outweigh the risks.

Projected Timeline

Activity	**Year 1**				**Year 2**				**Year 3**				**Year 4**				**Year 5**			
	Q1	**Q2**	**Q3**	**Q4**	**Q1**	**Q2**	**Q3**	**Q4**	**Q1**	**Q2**	**Q3**	**Q4**	**Q1**	**Q2**	**Q3**	**Q4**	**Q1**	**Q2**	**Q3**	**Q4**
Startup: IRB, hiring, supplies, etc.)																				
Aim 1 cohort recruitment period																				
Aim 1 follow-up and immune testing																				
Aim 2 treponemal biology																				
Aim 3 rapid test evaluation																				
Analysis, write-up and dissemination																				

Activity	**Year 6**				**Year 7**				**Year 8**				**Year 9**				**Year 10**			
	Q1	**Q2**	**Q3**	**Q4**	**Q1**	**Q2**	**Q3**	**Q4**	**Q1**	**Q2**	**Q3**	**Q4**	**Q1**	**Q2**	**Q3**	**Q4**	**Q1**	**Q2**	**Q3**	**Q4**
Amendment: IRB, hiring, supplies, etc.	X	X																		
Aim 1 cohort recruitment period			X	X	X	X	X	X	X	X										
Aim 1 follow-up			X	X	X	X	X	X	X	X		X	X							
Aim 4: host global patient transcriptome analysis					X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Aim 5: immune response characterization					X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Analysis, write-up and dissemination														X	X	X	X	X	X	X